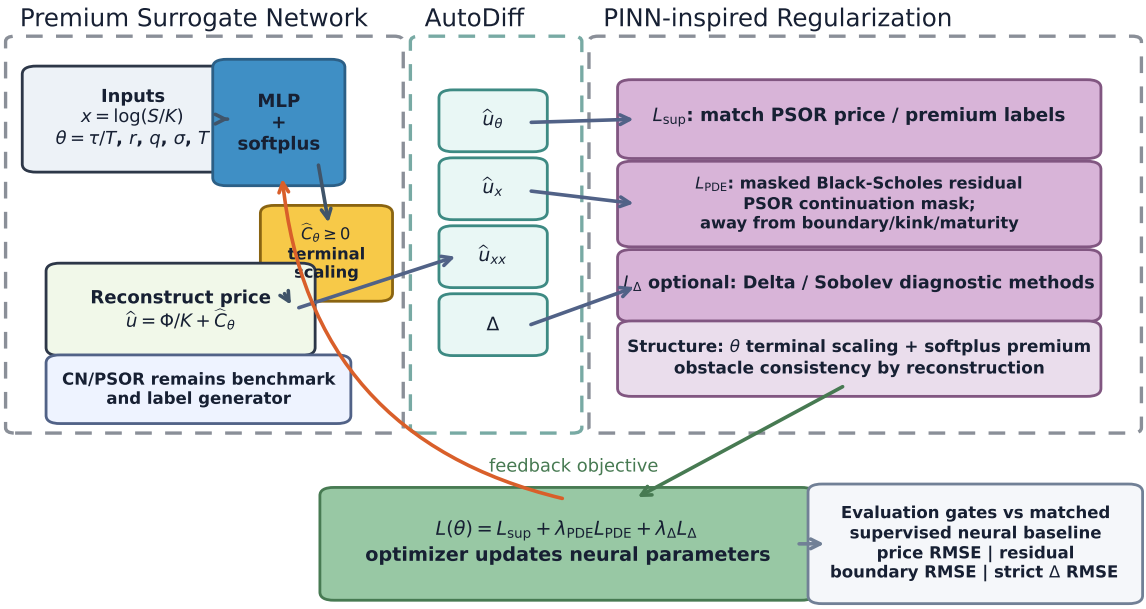
Project-specific PINN-inspired mechanism
Premium Surrogate Network
AutoDiff
PINN-inspired Regularization
Inputs
$x = \log(S/K)$
$\theta = \tau/T, r, q, \sigma, T$
MLP
+
softplus
$\widehat{C}_\theta \geq 0$
terminal
scaling
Reconstruct price
$\widehat{u} = \Phi/K + \widehat{C}_\theta$
CN/PSOR remains benchmark
and label generator
$\widehat{u}_\theta$
$\widehat{u}_x$
$\widehat{u}_{xx}$
$\Delta$
$L_{sup}$: match PSOR price / premium labels
$L_{PDE}$: masked Black-Scholes residual
PSOR continuation mask;
away from boundary/kink/maturity
$L_\Delta$ optional: Delta / Sobolev diagnostic methods
Structure: $\theta$ terminal scaling + softplus premium
obstacle consistency by reconstruction
feedback objective
$L(\theta) = L_{sup} + \lambda_{PDE} L_{PDE} + \lambda_\Delta L_\Delta$
optimizer updates neural parameters
Evaluation gates vs matched
supervised neural baseline
price RMSE | residual
boundary RMSE | strict Δ RMSE